Tips & Tricks with SAP Business Network for Logistics



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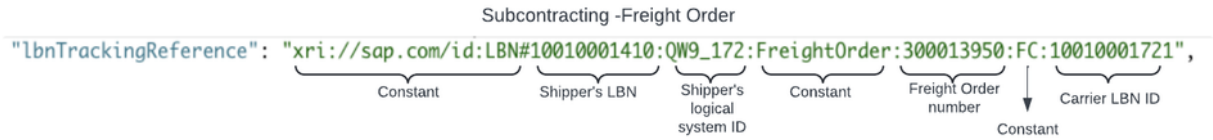
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Common queries/issues/know-hows in Business Network for Logistics and their resolution/explanation.



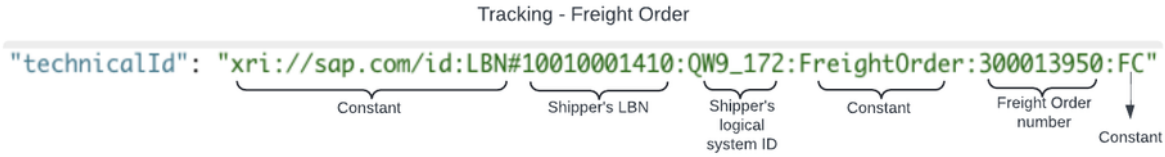
1. Technical IDs in APIs for Carrier Integration

Technical ID in Consumer API for Subcontracting



TechnicalID in consumer API for Subcontracting

Technical ID in Consumer API for Order Tracking



TechnicalID in consumer API for Tracking – Freight Order

- The constant at the end, **FC**, indicates that the shipper is using Freight Collaboration for Tracking process. Carriers have to make sure that they report events with the exact technical ID for successful reporting.

Technical ID in Consumer API for Booking Confirmation



TechnicalID in consumer API for Booking Confirmation

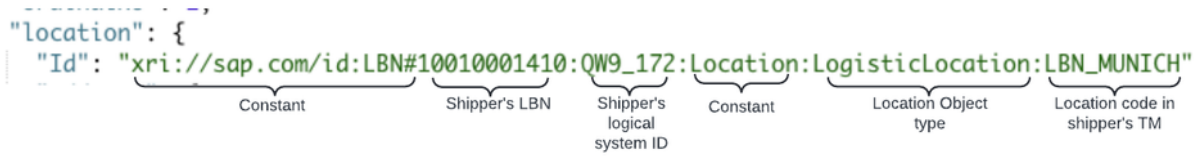
Technical ID in Consumer API for Order Tracking – Freight Booking



TechnicalID in consumer API for Tracking – Freight Booking

- Note that even though it is a Freight Booking, the technicalID uses Freight Order as the constant in the technicalId.

Location ID in Consumer API for Order Tracking



Location ID in consumer API for Tracking

- The Location Object Type constant can have different values depending on how a shipper configures the location master. Carriers should discuss with their shippers on possible values if relying on this field or if planning to report events with locationID.
- The Location code is shipper-specific code as mastered in their TM system.

Technical ID in Consumer API for Order Tracking – Global Track and Trace Shipment

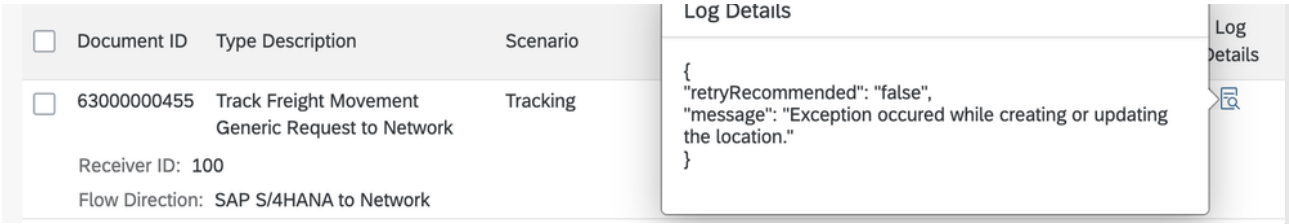


TechnicalID in consumer API for tracking – GTT

- The constant that indicates Tracking ID type differs for different tracked processes. The one above is commonly used as it is part of standard GTT model for Shipment Tracked process.
- If a customer uses a custom GTT model, this value could differ.
- The constant at the end, **GTT,** indicates that the shipper is using Global Track & Trace for the tracking process.

2. Exception occurred while creating or updating the location

Sometimes you see this error when a shipper initiates a tracking request. The error is not specific but does indicate a possible problem.



When a shipper initiates a tracking request, it is considered an A2A call and the system tries to update the location master data from TM into BNL. In this particular case, the shipper maintained a faulty UNLOCODE for the location. (3 characters as opposed to standard 5 characters).

So, if this error is seen, shippers should check the location master.

3. No matching stop found while reporting events

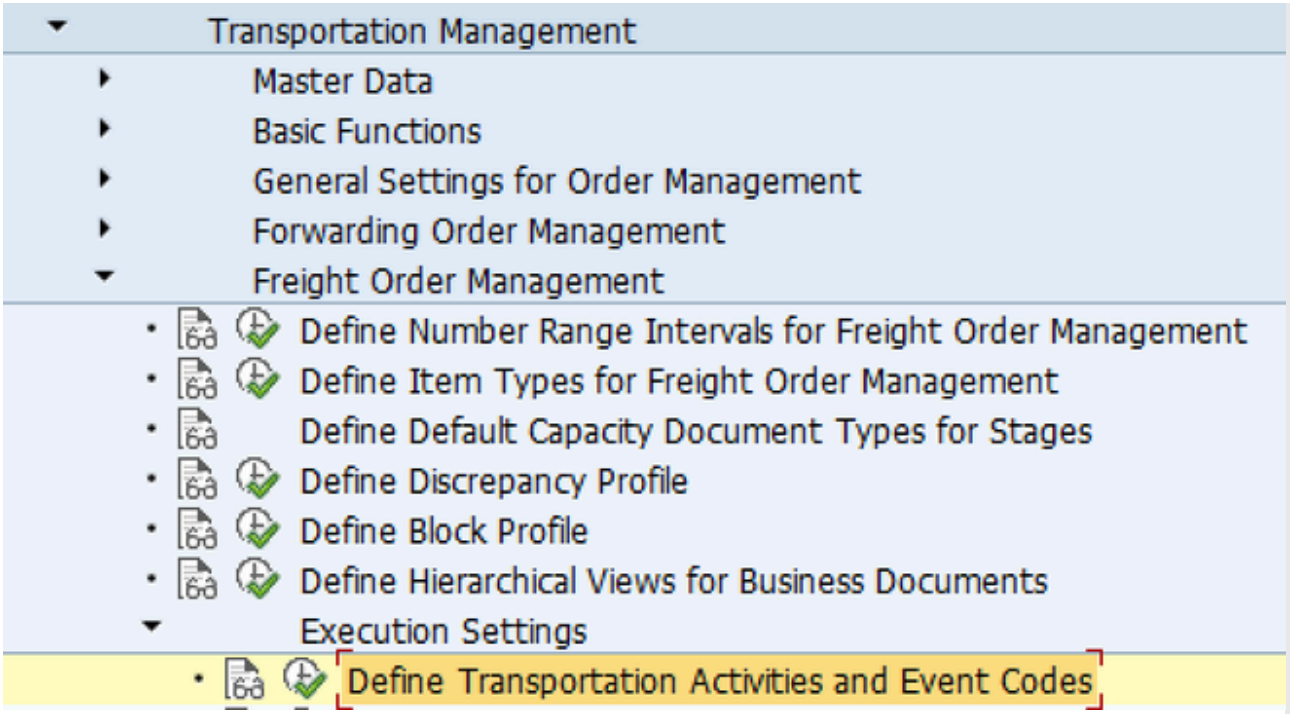
Carrier reports events via Tracking APIs and all the logs in BNL and CPI show it was successful. However, events are not seen reported against the Freight Order – Execution tab in TM.
On checking srt_moni you will notice some error logs:

Adapter Type: PLAINSOAP Message ID: A76816414365440A81D4F11C0748990F

Ex...	Name	Value
	-n0:Fault	
	->xmlns:n0	http://schemas.xmlsoap.org/soap/envelope/
	--faultcode	n0:Client
	--faultstring	Application exception occurred!
	-->lang	en
	--detail	
	---n1:StandardMessageFault	
	--->xmlns:prx	urn:sap.com:proxy:QW9:/1SAI/TAS0AA6B71EFAD45A977E80:816
	--->xmlns:n1	http://sap.com/xi/SAPGlobal20/Global
	----standard	
	-----faultText	Standard Message Fault
	-----faultDetail	
	------severity	warning
	------text	changeOrdinalNumberValue not specified in message = : serialization cannot be checked
	------id	545(/SCMTMS/TRQ)
	-----faultDetail	
	------severity	warning
	------text	Event DEPARTURE with expected date not reported; No matching stop found
	------id	008(/SCMTMS/INB_AGENT)
	-----faultDetail	
	------severity	error
	------text	Processing canceled; data received is not valid
	------id	002(/SCMTMS/INB_AGENT)

Log in srt_moni

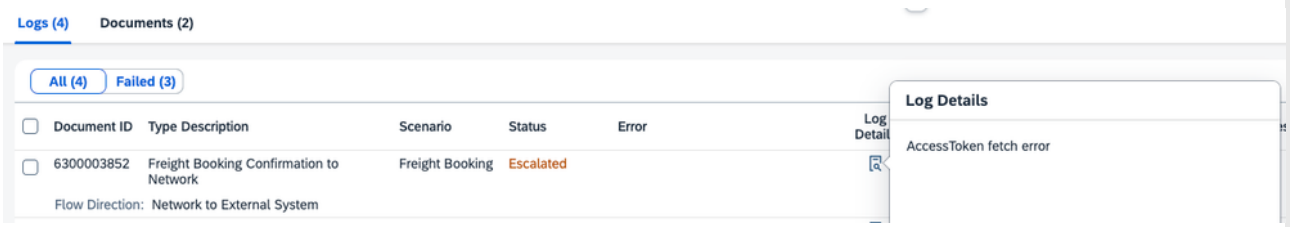
As the error indicates, the shipper would have configured mandatory stopIDs for one or more events. This configuration is available in:



- Sometimes shippers fail to convey this requirement to carriers that they have mandated stopIDs. BNL accepts events with or without stopID and reports it as planned or unplanned respectively.
- Carriers can choose to send stopIDs, locationID, or UNLOCODE/IATACode. If the carrier sends locationID or UNLOCODE/IATACode, BNL looks up the stopIDs and delegates it to TM. Usually carriers prefer to use UNLOCODE/IATACode as this is an industry-standard code and they do not have to deal with shipper-specific codes.

4. AccessToken fetch error

One of the scenarios where this error is seen is when a Network Partner is involved and is responding to a Freight Booking confirmation via the API. You will notice that the API call itself gives a 200 OK and also gives a JSON response, but the error will be seen in Manage Integration Logs app as below.

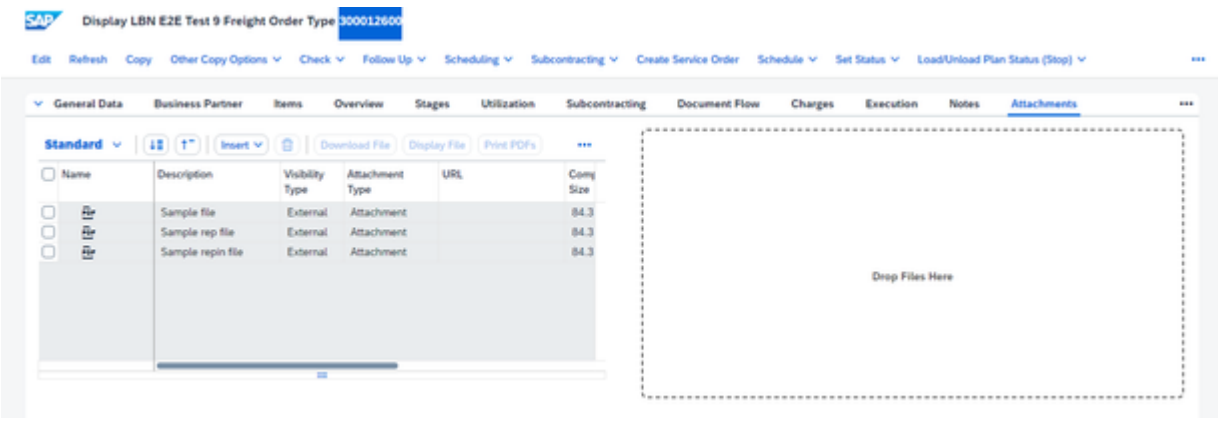


Freight booking API has fields, *documentRecipientParty, senderParty, and recipientParty.* The values in these fields differ in the consumer and provider calls, that the Network Partner makes. The error occurs when there is a mismatch in the value sin those fields resulting in token issues. To understand the usage, refer this blog.

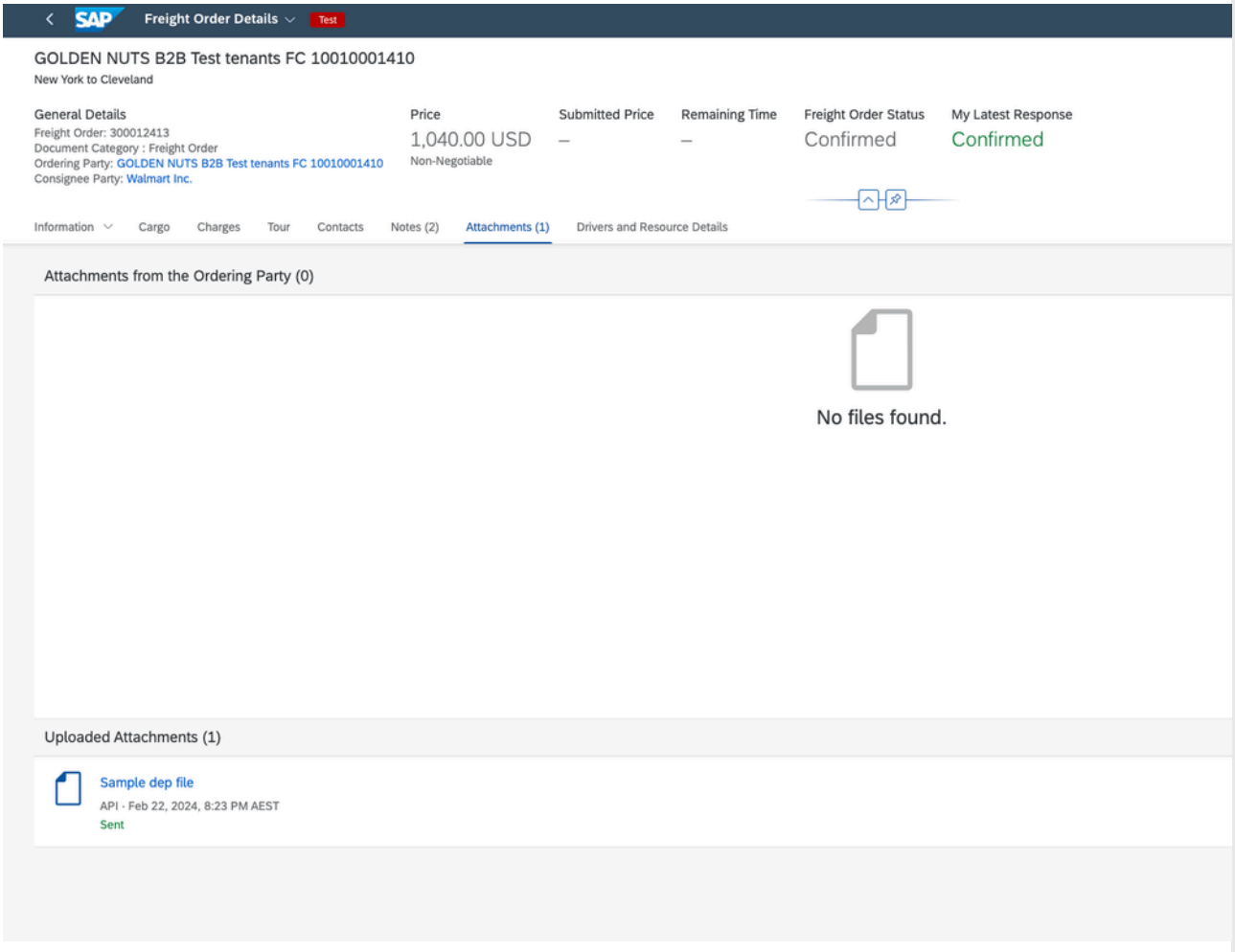
5. Attachments sent via APIs and application behavior.

General Attachments:

- Can be sent by carriers via their confirmation call or even updates to confirmation call. This can be done until the orders go into execution by the carrier (In certain versions of TM or S/4HANA confirmation updates can be sent even while the order is in execution).
- These attachments can be seen by the shipper in the attachment section against the orders in TM.



- Carriers can also see these attachments in FOFC (Freight Order for Confirmation) app in BNL.

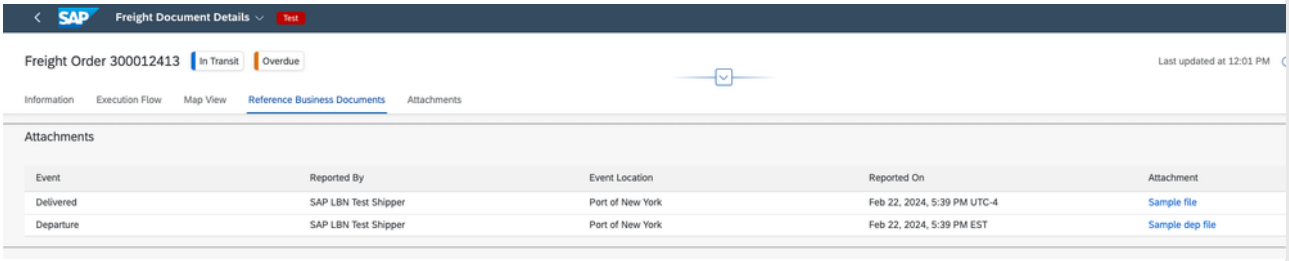


Attachments in Freight Order for Confirmation app

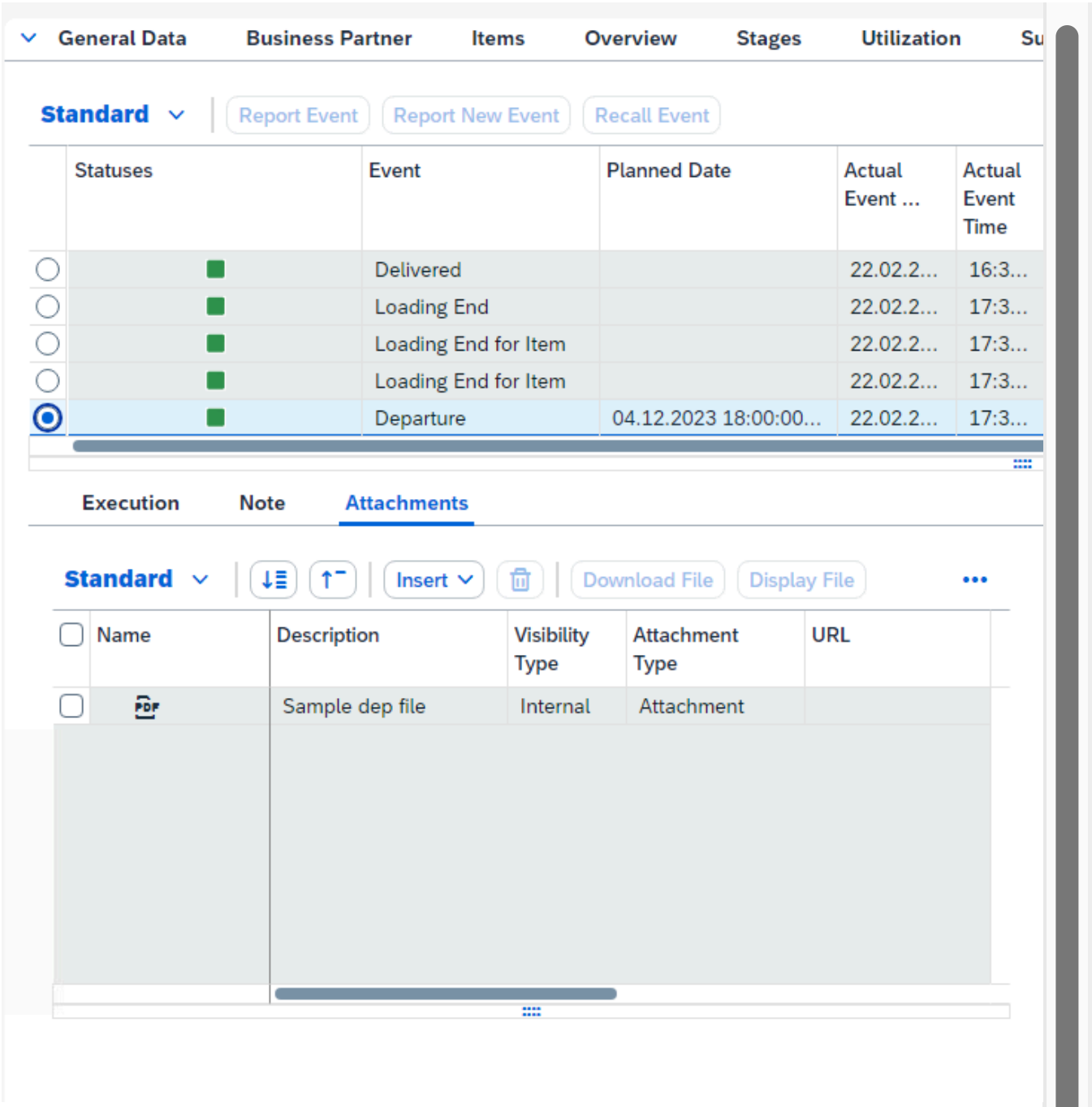
- If any updates to confirmation are made while the order is in execution, the attachments are seen by the carrier in BNL, but they will not get processed by the shipper and cannot be seen in TM (only on certain versions of TM).

Event Attachments:

- These are sent by carriers when reporting events via API.
- If attachments are sent against planned events, these are seen by the carrier in FOFC app.
    - Shippers can also see these attachments in the TFM (Track Freight Movement) app in BNL and also in TM in the attachment section of each event. (Not in the general attachment section) (egs: DEPARTURE planned event)



Attachments shown in Track Freight movement app



- If attachments are sent against unplanned events, these are seen by the shipper in the TFM app in BNL and also in TM in the attachment section of the event.
- The carrier cannot see the attachment sent for unplanned events in FOFC BNL app.

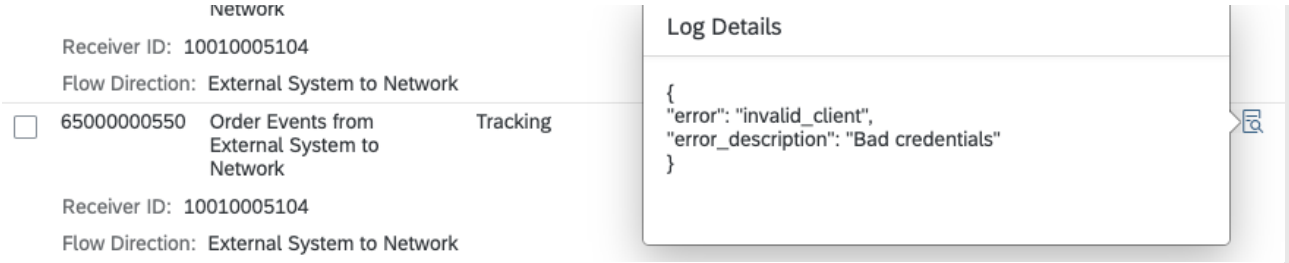
6. Timezones and event reporting

Scenario	UI/API	FC	GTT
Location has timeZone and event reported without timeZone (API)	API	UTC reported time is converted to location Timezone	UTC reported time converted to users local timezone.

Location has no timeZone and event reported without timezone (API)	API	Cannot create Locations without Timezone.	UTC reported time converted to users local timezone.
Location has TimeZone & event reported with a different timeZone (API)	API	UTC reported time is converted to location Timezone	UTC reported time converted to users local timezone.
Location has no TimeZone & event reported with a different timeZone (API)	API	Cannot create Locations without Timezone.	UTC reported time converted to users local timezone.
Location has TimeZone & event reported with a different timeZone (UI)	UI	UTC reported time is converted to location Timezone	The time reported is considered to be for the timezone given and the user sees converted local Time
Location has no TimeZone & event reported with a different timeZone (UI)	UI	Cannot create Locations without Timezone.	The time reported is considered to be for the timezone given and the user sees converted local Time
Unplanned Events with timeZone	API	Converts it to the time based on the timeZone sent and shows that timezone as well in the UI.	UTC reported time converted to users local timezone.

7. invalid_client, Bad Credentials error while reporting events via API

Sometimes when a carrier reports events the below error is seen. The credentials used are also correct.



Error in MIL app

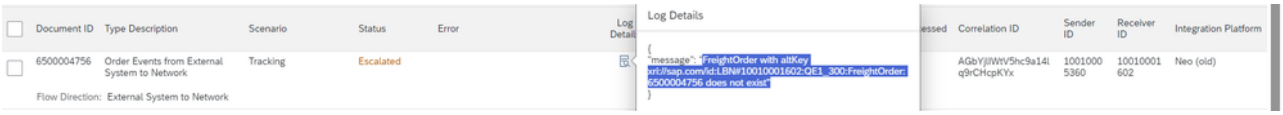
This happens when the shipper is subscribed to FC but the carrier is reporting event with the technicalId suffixed with GTT or vice versa.

egs: *"xri://sap.com/id: LBN#10010005000:BS_SAP:FreightBooking:65000000550:GTT".*

To resolve this, the carrier has to make sure the right suffix is sent after discussing with the shipper.

8. FreightOrder with altKey does not exist error

While reporting events, carriers sometimes see this error in the MIL app. The call itself to report events will respond with a 200 OK.



This happens when the Order is not set to *"Ready for Transportation Execution"* by the shipper. To resolve this, ask the shipper to change the status of the Order, following which events can be reported.

9. JSON parse error: Cannot deserialize value of type com.sap.lbn.trackingservice.domain.input.generic.TransportationMode from String "04": not one of the values accepted for Enum class: [01, 03]

This error is seen in MIL(Manage Integration Logs) app when a carrier is trying to report events via API.